

EXP NO:	MINI-PROJECT 1 FINANCIAL DATA ANALYTICS DASHBOARD
DATE:	

AIM:

To analyze and visualize financial data using Microsoft Power BI and Tableau by connecting and importing data from SQL, performing data cleaning and transformation, creating meaningful visualizations and dashboards, applying calculations and analytical techniques, and developing interactive reports to identify important financial trends, patterns, and insights.

PROBLEM STATEMENT:

To analyze financial data containing information about sales, revenue, costs, profits, products, regions, customers, and sales channels by using SQL for data management and Power BI and Tableau for analysis and visualization. The project aims to clean and transform the data, perform relevant calculations, and develop interactive dashboards to identify financial trends, profitability, regional and product performance, and customer patterns, thereby supporting effective business decision-making.

DATA DESCRIPTION:

The Financial Data Analytics dataset contains 589 transaction records representing financial and sales activities across different regions, departments, products, customers, and sales channels. The dataset includes both categorical and numerical attributes that can be used for statistical analysis, visualization, and business intelligence.

Category	Attributes
Transaction Details	Transaction ID, Date, Year, Month
Geographical Details	Region, State, City
Product Details	Department, Product
Customer Details	Customer Type
Sales Details	Sales Channel, Payment Method, Units Sold
Financial Details	Unit Price, Revenue, Discount Percentage, Discount Amount, Cost, Profit, Profit Margin
Customer Feedback	Customer Rating

DATA SOURCE:

The financial dataset used for this project is derived from a Kaggle financial dataset. The dataset was adapted and structured for financial data analysis by organizing information

related to transactions, products, customers, regions, sales channels, revenue, costs, discounts, and profits.

Source: Kaggle – Financial Dataset

Data Type: Structured financial transaction data

Records: 589 transactions

Format: CSV/Excel, imported into MySQL for database management and subsequently connected to Power BI and Tableau.

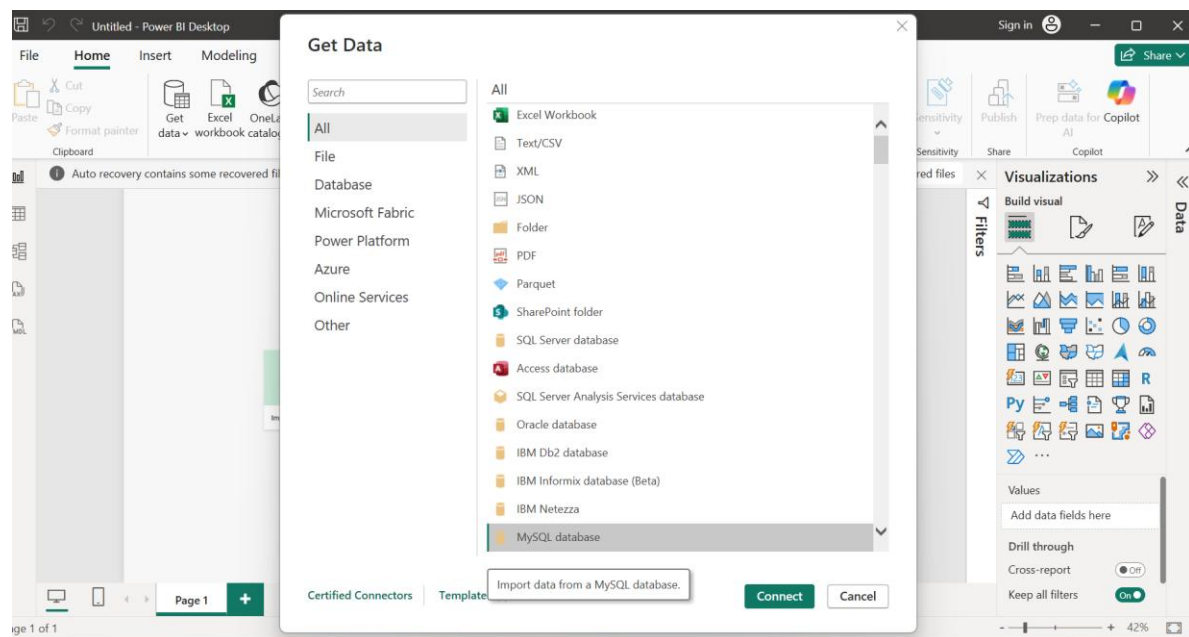
POWER BI DASHBOARD:

1)IMPORTING THE DATA FROM MYSQL:

Algorithm:

- Open Power BI Desktop.
- Select **Get Data** → **MySQL Database**.
- Enter the MySQL server and database details.
- Select the required financial table.
- Click **Transform Data** to import the data into Power Query.

Output:



DataQueriesInsertCalculationsSensitivity

ains some recovered files that haven't been opened.

View recovered files

MySQL database

Serverlocalhost:3306

Databaseanalytics_project

Advanced options

OKCancel

Power BI

Insert

Get data

very contains

Navigator

Display Options

localhost:3306: analytics_project [3]

☒ analytics_project.financial_data

☐ analytics_project.healthcare_data

☐ analytics_project.healthcare_data_analytics

analytics_project.financial_data

Transaction_ID	Date	Year	Month	Region	State
TXN00001	2025-10-16	2025	October	North	Delhi
TXN00002	2024-01-31	2024	January	North	Punjab
TXN00003	2026-02-16	2026	February	South	Andhra Pr
TXN00004	2026-04-06	2026	April	North	Uttar Prad
TXN00005	2026-03-02	2026	March	East	West Beng
TXN00006	2025-11-30	2025	November	North	Rajasthan
TXN00007	2024-11-28	2024	November	North	Punjab
TXN00008	2026-06-24	2026	June	West	Gujarat
TXN00009	2024-02-18	2024	February	North	Punjab
TXN00010	2026-08-23	2026	August	North	Delhi
TXN00011	2024-09-26	2024	September	South	Telangana
TXN00012	2025-05-15	2025	May	North	Delhi
TXN00013	2025-06-29	2025	June	South	Karnataka
TXN00014	2025-12-07	2025	December	South	Kerala
TXN00015	2024-08-13	2024	August	North	Delhi
TXN00016	2024-05-15	2024	May	West	Gujarat
TXN00017	2025-10-31	2025	October	North	Delhi
TXN00018	2026-06-14	2026	June	North	Uttar Prad
TXN00019	2024-03-01	2024	March	South	Andhra Pr
TXN00020	2024-07-13	2024	July	East	Odisha
TXN00021	2024-02-28	2024	February	North	Punjab
TXN00022	2025-11-04	2025	November	East	Bihar

Select Related Tables

LoadTransform DataCancel

File Home Help **Tools**

Name analytics_project fi...

Manage relationships
New measure
Quick measure column
New table
Mark as date table
Calendars

Transaction_ID	Date	Year	Month	Region	State	City	Department	Product	Customer_Type	Sales Char
TXN00001	2025-10-16	2025	October	North	Delhi	Rohini	Grocery	Sugar 1kg	New	Online
TXN00002	2024-01-31	2024	January	North	Punjab	Amritsar	Sports	Fitness Tracker	New	Online
TXN00003	2026-02-16	2026	February	South	Andhra Pradesh	Vijayawada	Grocery	Wheat Flour 5kg	New	Offline
TXN00004	2026-04-06	2026	April	North	Uttar Pradesh	Noida	Electronics	Digital Camera	New	Offline
TXN00005	2026-03-02	2026	March	East	West Bengal	Kolkata	Electronics	Digital Camera	Returning	Offline
TXN00006	2025-11-30	2025	November	North	Rajasthan	Jodhpur	Clothing	Casual Sneakers	New	Online
TXN00007	2024-11-28	2024	November	North	Punjab	Amritsar	Grocery	Spices Combo Pack	Returning	Online
TXN00008	2026-06-24	2026	June	West	Gujarat	Surat	Clothing	Formal Shirt	Returning	Offline
TXN00009	2024-02-18	2024	February	North	Punjab	Jalandhar	Clothing	Ethnic Saree	New	Offline
TXN00010	2026-08-23	2026	August	North	Delhi	Rohini	Sports	Running Shoes	Returning	Online
TXN00011	2024-09-26	2024	September	South	Telangana	Hyderabad	Grocery	Dry Fruits Pack	New	Online
TXN00012	2025-05-15	2025	May	North	Delhi	Dwarka	Grocery	Sugar 1kg	New	Online
TXN00013	2025-06-29	2025	June	South	Karnataka	Mangaluru	Furniture	TV Unit	New	Offline
TXN00014	2025-12-07	2025	December	South	Kerala	Thiruvananthapuram	Grocery	Breakfast Cereal	Returning	Online
TXN00015	2024-08-13	2024	August	North	Delhi	Rohini	Electronics	Smart TV	New	Online
TXN00016	2024-05-15	2024	May	West	Gujarat	Surat	Furniture	Study Table	New	Online
TXN00017	2025-10-31	2025	October	North	Delhi	Dwarka	Grocery	Cooking Oil 1L	New	Online
TXN00018	2026-06-14	2026	June	North	Uttar Pradesh	Noida	Electronics	Smartphone	New	Online
TXN00019	2024-03-01	2024	March	South	Andhra Pradesh	Visakhapatnam	Furniture	Bookshelf	Returning	Offline
TXN00020	2024-07-13	2024	July	East	Odisha	Bhubaneswar	Sports	Skipping Rope	New	Offline

Table: analytics project financial data (589 rows)

The screenshot displays the Power Query Editor window. The ribbon at the top includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The main area shows a data table with columns Transaction_ID, Date, and Year. The Date column is selected, and a context menu is open, showing various data type options. The bottom status bar indicates '21 COLUMNS, 589 ROWS' and 'Column profiling based on top 1000 rows'.

2)Removing empty rows:

The screenshot shows the Power Query Editor interface. The formula bar contains the query: `= Table.TransformColumnTypes(analytics_project_financial_data,{{"Date", type date}})`. The data table has columns: Transaction_ID, Date, Year, and Month. The 'Remove Rows' menu is open, and 'Remove Blank Rows' is selected. The 'APPLIED STEPS' pane on the right shows 'Source', 'Navigation', and 'Changed Type'.

Transaction_ID	Date	Year	Month
TXN00001	16-10-2025	2025	October
TXN00002	31-01-2024	2024	January
TXN00003	16-02-2026	2026	February
TXN00004	06-04-2026	2026	April
TXN00005	02-03-2026	2026	March
TXN00006	30-11-2025	2025	November
TXN00007	28-11-2024	2024	November
TXN00008	24-06-2026	2026	June
TXN00009	18-02-2024	2024	February
TXN00010	23-08-2026	2026	August
TXN00011	26-09-2024	2024	September
TXN00012	15-05-2025	2025	May
TXN00013	29-06-2025	2025	June
TXN00014	07-12-2025	2025	December

3)Removing duplicate rows:

The screenshot shows the Power Query Editor interface. The formula bar contains the query: `= Table.SelectRows("#\"Changed Type\"", each not List.IsEmpty)`. The 'Remove Rows' menu is open, and 'Remove Duplicates' is selected. The 'APPLIED STEPS' pane on the right shows 'Source', 'Navigation', 'Changed Type', and 'Remove Duplicates'.

Transaction_ID	Date	Year	Month
TXN00001	16-10-2025	2025	October
TXN00002	31-01-2024	2024	January
TXN00003	16-02-2026	2026	February
TXN00004	06-04-2026	2026	April
TXN00005	02-03-2026	2026	March
TXN00006	30-11-2025	2025	November
TXN00007	28-11-2024	2024	November
TXN00008	24-06-2026	2026	June
TXN00009	18-02-2024	2024	February
TXN00010	23-08-2026	2026	August
TXN00011	26-09-2024	2024	September
TXN00012	15-05-2025	2025	May
TXN00013	29-06-2025	2025	June
TXN00014	07-12-2025	2025	December

4)Applying all the changes and closing:

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By

Apply Close

analytics_project financia...

= Table.Distinct(#"Removed Blank Rows", {"Date"})

	Transaction_ID	Date	Year	Month
1	TXN00001	16-10-2025	2025	October
2	TXN00002	31-01-2024	2024	January
3	TXN00003	16-02-2026	2026	February
4	TXN00004	06-04-2026	2026	April
5	TXN00005	02-03-2026	2026	March
6	TXN00006	30-11-2025	2025	November
7	TXN00007	28-11-2024	2024	November
8	TXN00008	24-06-2026	2026	June
9	TXN00009	18-02-2024	2024	February
10	TXN00010	23-08-2026	2026	August
11	TXN00011	26-09-2024	2024	September
12	TXN00012	15-05-2025	2025	May
13	TXN00013	29-06-2025	2025	June
14	TXN00014	07-12-2025	2025	December

3)DATA MODELLING:

Algorithm

1. Open Model view in Power BI.
2. Check the imported financial table.
3. Create a Date table using DAX.
4. Establish a relationship between the Date table and financial data.
5. Verify the relationship.

Output:

1)Creating new date table

File Home Help

Get data Excel workbook OneLake catalog SQL Server Recent sources Transform data Refresh Manage relationships New measure column New table Calculation group New parameter Manage roles

Clipboard

1 DataTable = CALENDAR(DATE(2023,1,1), DATE(2025,12,31))

analytics_project financia...

- City
- Cost
- Customer_Rating
- Customer_Type
- Date
- Department
- Discount_Amount
- Discount_Percentage
- Month

Table

Column

2)Setting the relationship between the table and setting the relationship as one-to-many and cross-filter direction as single:

New relationship

Select tables and columns that are related.

From table

DateTable

Date

01-01-2023 0...
02-01-2023 0...
03-01-2023 0...

To table

analytics_project financial_data

City	Cost	Customer_Rat...	Customer_Type	Date	Department	Discount_
Rohini	11428.85	2.69	New	16 October 2...	Grocery	3967.79
Amritsar	31052.71	4.48	New	31 January 20...	Sports	9804.2
Vijayawada	1174.81	3.41	New	16 February 2...	Grocery	146.08

Cardinality

One to many (1:*)

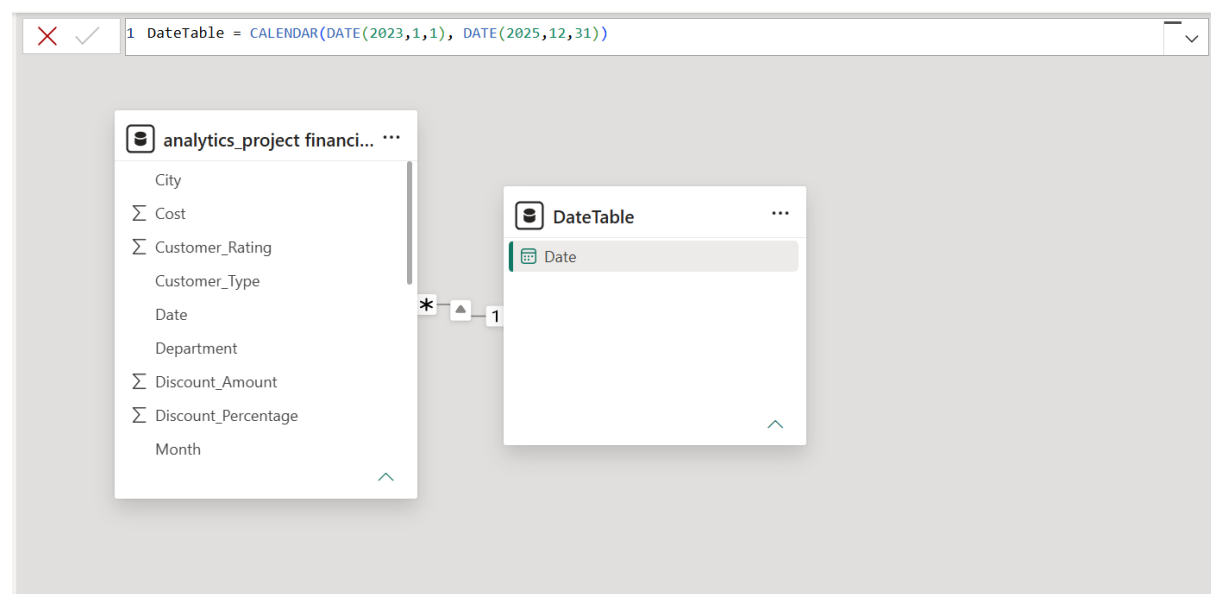
Cross-filter direction

Single

☒ Make this relationship active ☐ Apply security filter in both directions

Save Cancel

3)Create data model



4)CREATE CALCULATED COLUMNS:

Algorithm

1. Open the Data view.
2. Select the financial table.
3. Create a calculated column using DAX.
4. Create categories based on financial values.
5. Verify the new columns.

Output

1)Create profit category

s_Sold	Unit Price	Revenue	Discount_Percentage	Discount_Amount	Cost	Profit	Profit_Margin	Customer_Rating	Profit Category
22	901.77	19838.94	20	3967.79	11428.85	4442.3	22.39	2.69	Profit
14	3501.5	49021	20	9804.2	31052.71	8164.09	16.65	4.48	Profit
3	486.92	1460.76	10	146.08	1174.81	139.87	9.58	3.41	Profit
12	35275.1	423301.2	25	105825.3	241648.49	75827.41	17.91	3.65	Profit
6	23155.59	138933.54	5	6946.68	104346.39	27640.47	19.89	3.81	Profit
13	1433.05	18629.65	25	4657.41	14092.37	-120.13	-0.64	1.88	Loss
19	1057.82	20098.58	10	2009.86	12336.27	5752.45	28.62	3	Profit
13	1819.75	23656.75	5	1182.84	16627.33	5846.58	24.71	1.36	Profit
20	4483.71	89674.2	20	17934.84	56084.42	15654.94	17.46	3.21	Profit
14	2582.83	36159.62	0	0	30234.68	5924.94	16.39	4.5	Profit
25	235.78	5894.5	20	1178.9	4866.19	-150.59	-2.55	3.4	Loss
3	148.5	445.5	15	66.83	354.09	24.58	5.52	4.91	Profit
20	20194.93	403898.6	5	20194.93	334700.29	49003.38	12.13	4.02	Profit
8	123.63	989.04	0	0	718.53	270.51	27.35	1.92	Profit
3	31578.13	94734.39	10	9473.44	71116.48	14144.47	14.93	1.86	Profit
12	20213.88	242566.56	15	36384.98	196272.98	9908.6	4.08	1.22	Profit
18	565.91	10186.38	15	1527.96	6163.23	2495.19	24.5	2.85	Profit
6	25275.22	151651.32	15	22747.7	93132.74	35770.88	23.59	2.6	Profit
23	43164.66	992787.18	20	198557.44	743162.15	51067.59	5.14	2.95	Profit
2	8866.62	17733.24	20	3546.65	14644.4	-457.81	-2.58	3.12	Loss

2)Creating revenue category

Revenue	Discount_Percentage	Discount_Amount	Cost	Profit	Profit_Margin	Customer_Rating	Profit Category	Revenue Category
19838.94	20	3967.79	11428.85	4442.3	22.39	2.69	Profit	High Revenue
49021	20	9804.2	31052.71	8164.09	16.65	4.48	Profit	High Revenue
1460.76	10	146.08	1174.81	139.87	9.58	3.41	Profit	Low Revenue
423301.2	25	105825.3	241648.49	75827.41	17.91	3.65	Profit	High Revenue
138933.54	5	6946.68	104346.39	27640.47	19.89	3.81	Profit	High Revenue
18629.65	25	4657.41	14092.37	-120.13	-0.64	1.88	Loss	High Revenue
20098.58	10	2009.86	12336.27	5752.45	28.62	3	Profit	High Revenue
23656.75	5	1182.84	16627.33	5846.58	24.71	1.36	Profit	High Revenue
89674.2	20	17934.84	56084.42	15654.94	17.46	3.21	Profit	High Revenue
36159.62	0	0	30234.68	5924.94	16.39	4.5	Profit	High Revenue
5894.5	20	1178.9	4866.19	-150.59	-2.55	3.4	Loss	Low Revenue
445.5	15	66.83	354.09	24.58	5.52	4.91	Profit	Low Revenue
403898.6	5	20194.93	334700.29	49003.38	12.13	4.02	Profit	High Revenue
989.04	0	0	718.53	270.51	27.35	1.92	Profit	Low Revenue
94734.39	10	9473.44	71116.48	14144.47	14.93	1.86	Profit	High Revenue
242566.56	15	36384.98	196272.98	9908.6	4.08	1.22	Profit	High Revenue
10186.38	15	1527.96	6163.23	2495.19	24.5	2.85	Profit	High Revenue
151651.32	15	22747.7	93132.74	35770.88	23.59	2.6	Profit	High Revenue
992787.18	20	198557.44	743162.15	51067.59	5.14	2.95	Profit	High Revenue
17733.24	20	3546.65	14644.4	-457.81	-2.58	3.12	Loss	High Revenue

5)CREATE MEASURES USING DAX

Algorithm

1. Select the financial table.

2. Create a new measure using DAX.
3. Calculate revenue, cost, profit, and units sold.
4. Create measures for average revenue and customer rating.
5. Verify the measures.

Output

1)Total revenue measure:

File	Home	Insert	Modeling	View	Optimize	Help	Table tools	Measure tools
Name: Total Revenue		Format: General		Data category: Uncategorized		New measure Quick measure		
Home table: analytics_project fi...		Formatting: \$ % , -00 Auto		Properties		Calculations		
1 Total Revenue = SUM('analytics_project financial_data'[Revenue])								

2)Total cost measure:

File	Home	Insert	Modeling	View	Optimize	Help	Table tools	Measure tools
Name: Total Cost		Format: General		Data category: Uncategorized		New measure Quick measure		
Home table: analytics_project fi...		Formatting: \$ % , -00 Auto		Properties		Calculations		
1 Total Cost = SUM('analytics_project financial_data'[Cost])								

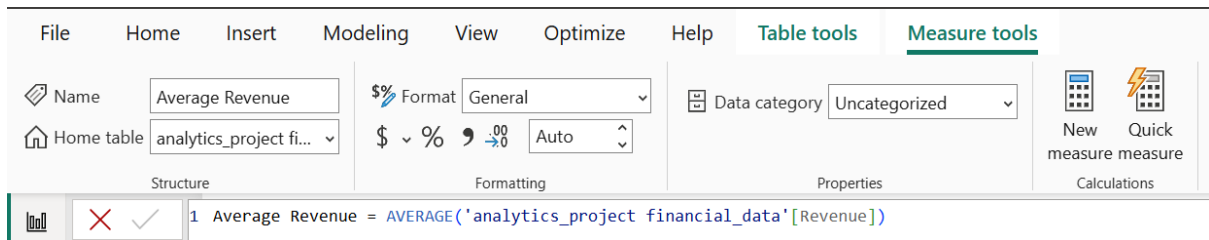
3)Total profit measure:

File	Home	Insert	Modeling	View	Optimize	Help	Table tools	Measure tools
Name: Total Profit		Format: General		Data category: Uncategorized		New measure Quick measure		
Home table: analytics_project fi...		Formatting: \$ % , -00 Auto		Properties		Calculations		
1 Total Profit = SUM('analytics_project financial_data'[Profit])								

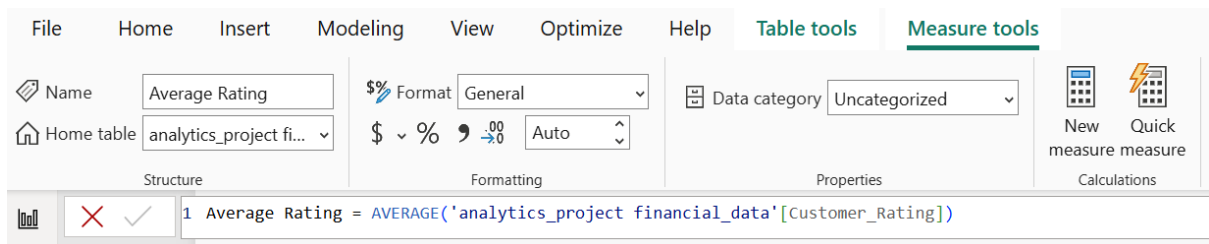
4) Total units sold measure:

File	Home	Insert	Modeling	View	Optimize	Help	Table tools	Measure tools
Name: Total Units Sold		Format: General		Data category: Uncategorized		New measure Quick measure		
Home table: analytics_project fi...		Formatting: \$ % , -00 Auto		Properties		Calculations		
1 Total Units Sold = SUM('analytics_project financial_data'[Units_Sold])								

5)Average revenue measure:



6)Average customer rating measure:

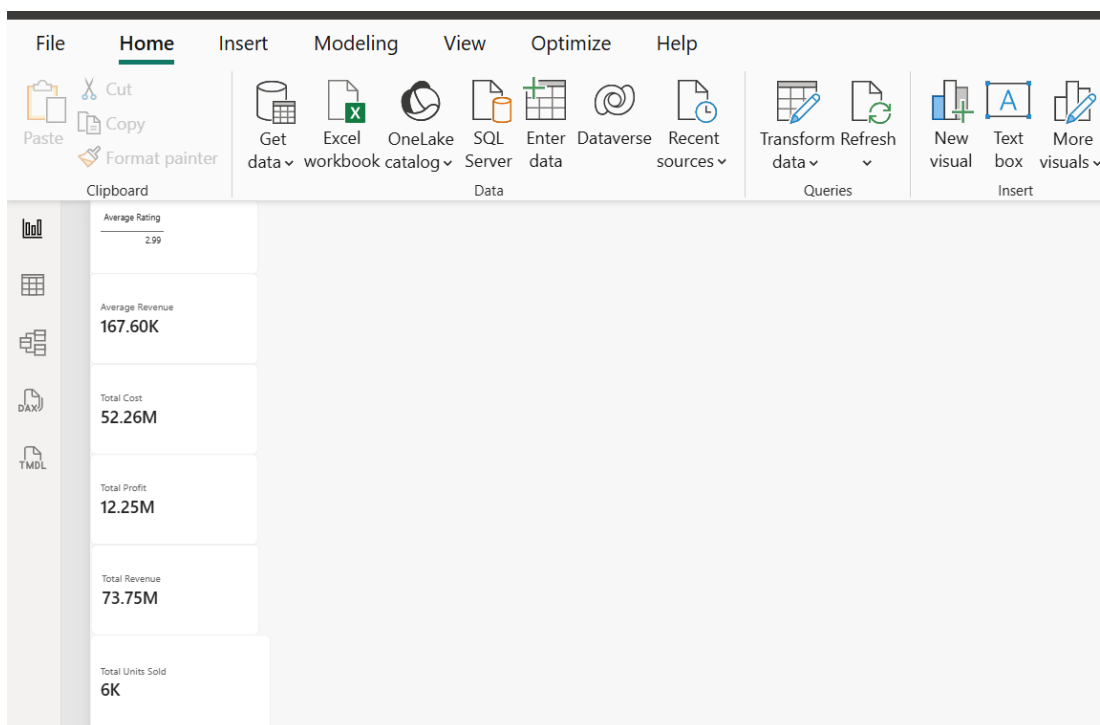


6)CREATING KPI CARDS:

Algorithm

1. Open Report View.
2. Insert a Card visual.
3. Add a financial measure to the card.
4. Repeat for the required measures.
5. Format the cards.

Output:



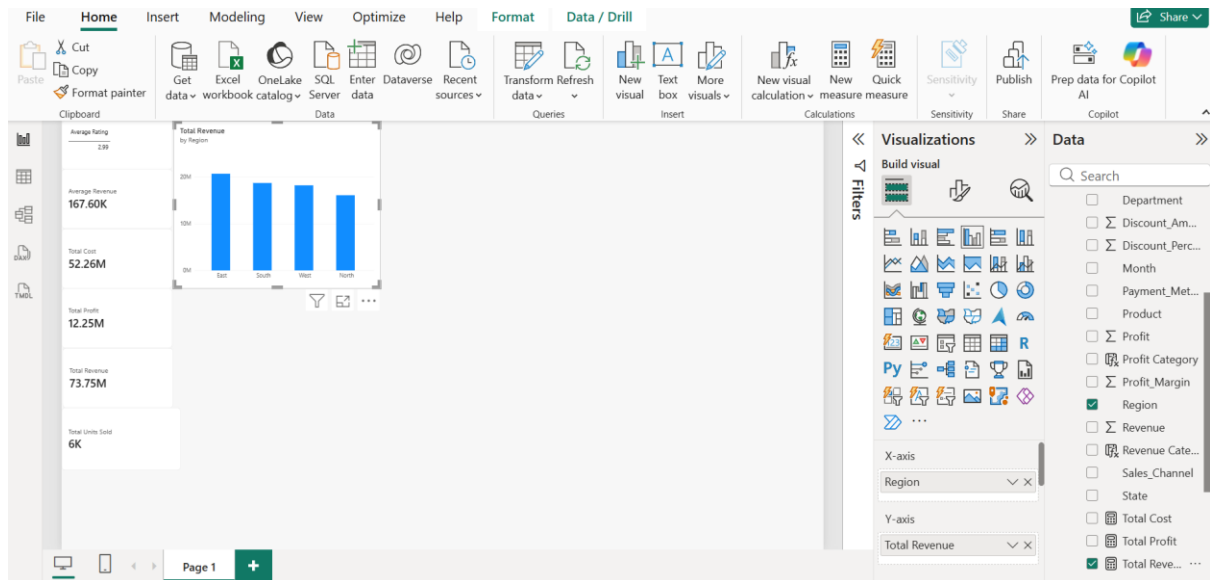
7) CREATING VISUALIZATIONS

Algorithm

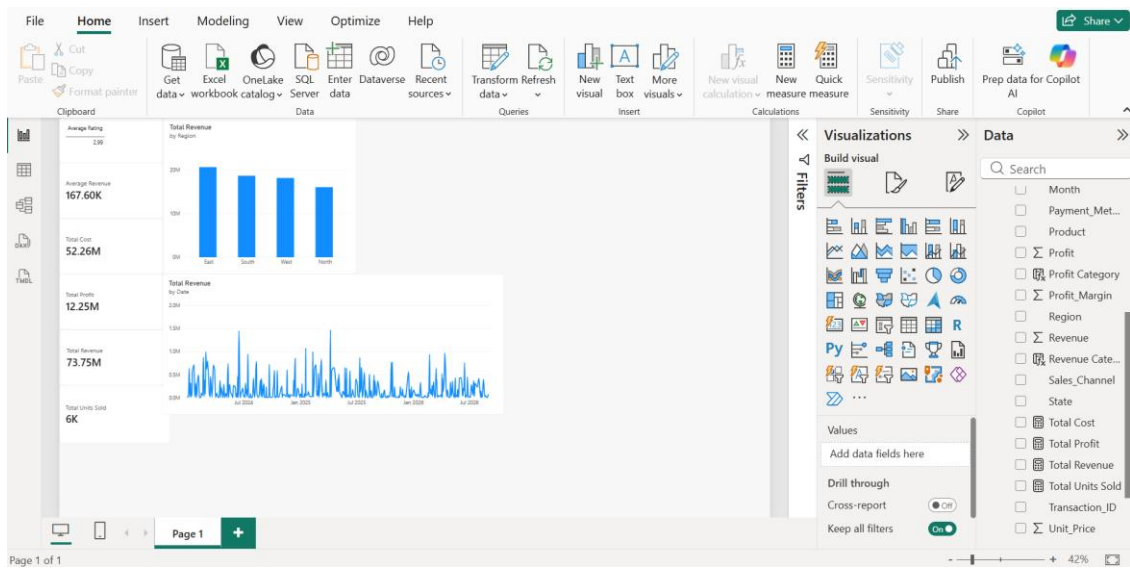
1. Create a column chart for regional revenue.
2. Create a line chart for monthly revenue trends.
3. Create a donut chart for sales-channel revenue.
4. Create a bar chart for product profitability.
5. Create a scatter chart to analyze revenue and customer ratings.
6. Format the visuals for a clear and attractive dashboard.

Output

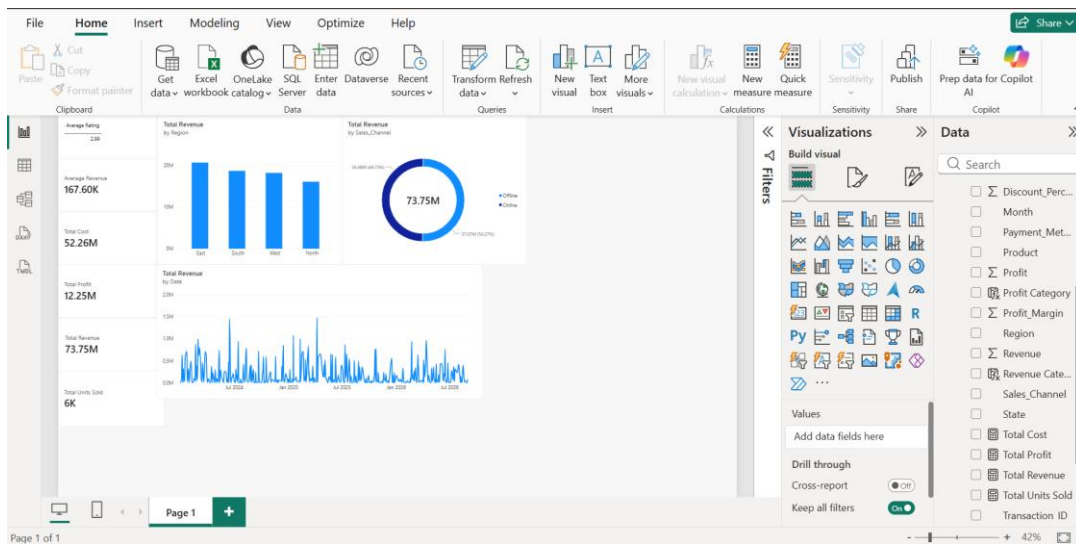
1) Revenue by region – column chart



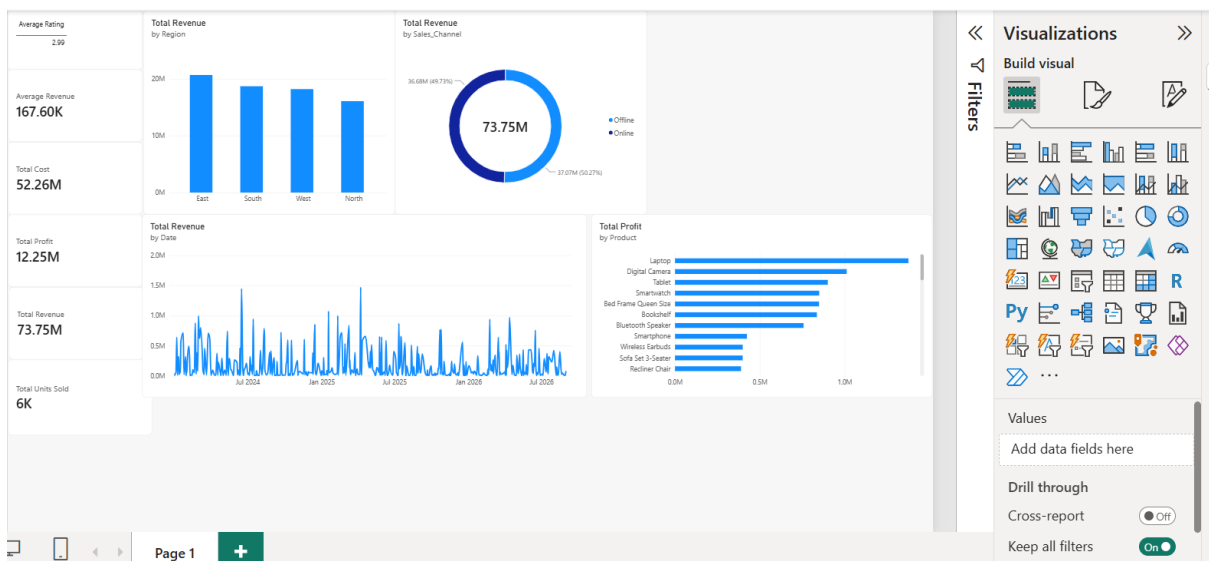
2) Monthly revenue trend



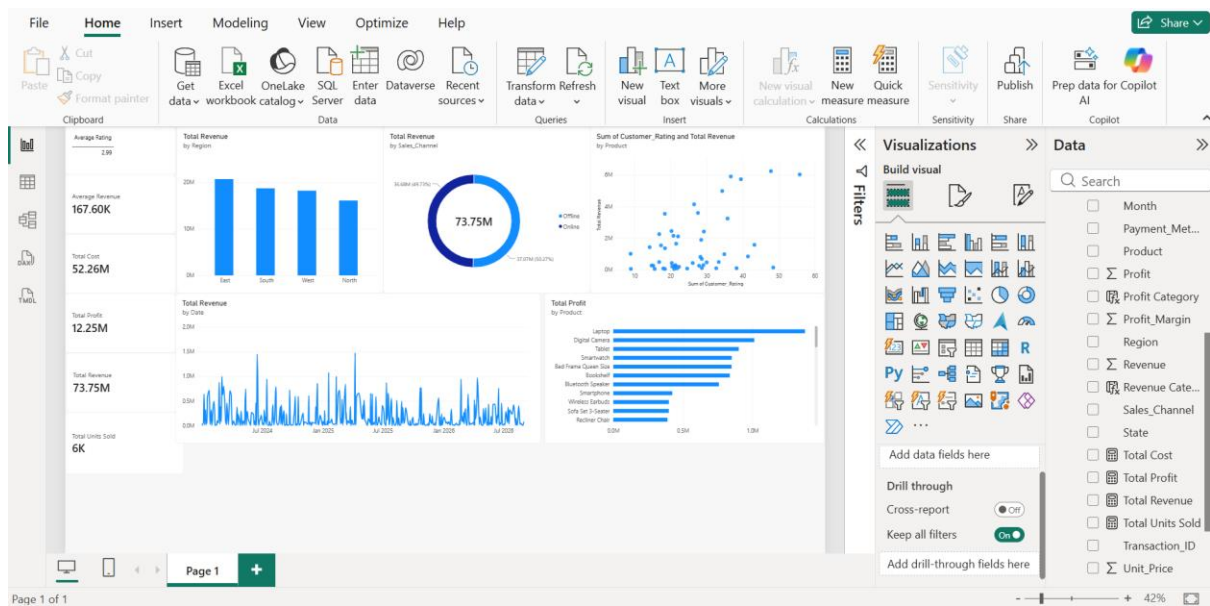
3)Online vs Offline revenue



4)Profit by product bar chart



5)Revenue vs customer rating



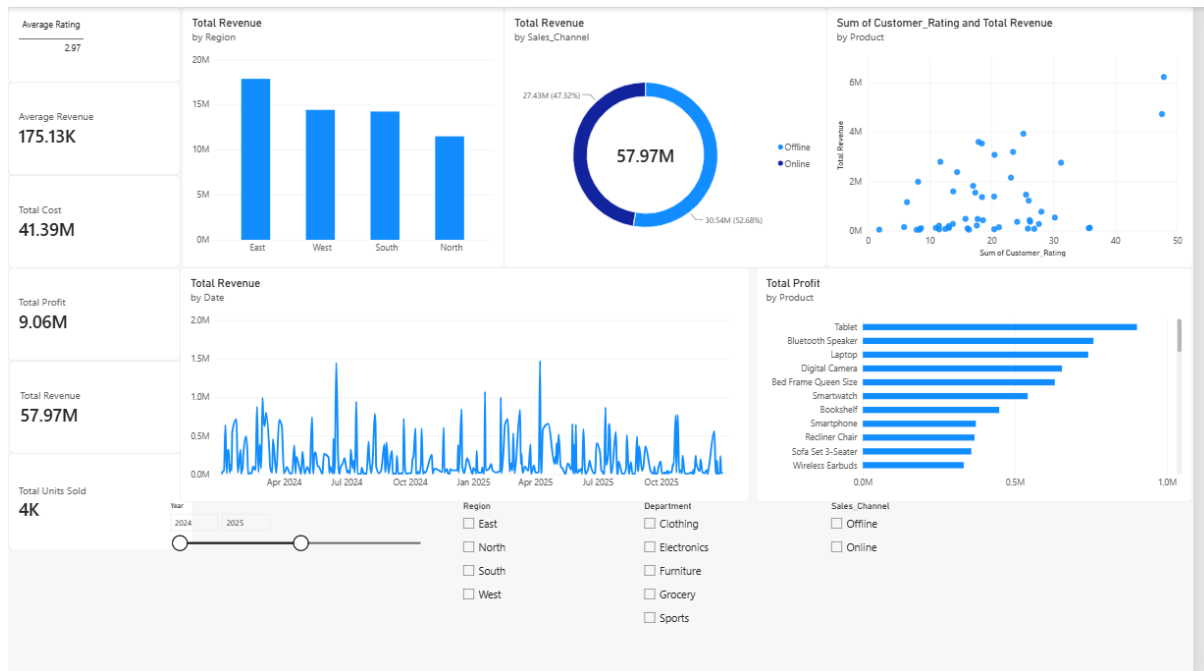
8)CREATING SLICERS AND FILTERS

Algorithm

1. Open the financial dashboard in Report View.
2. Insert a Slicer visual from the Visualizations pane.
3. Add the Year field to the slicer to enable year-wise filtering.
4. Create additional slicers using Region, Department, and Sales_Channel.
5. Arrange and format the slicers neatly on the dashboard.
6. Verify that selecting a value in a slicer filters the KPI cards and other visualizations.
7. Use the Edit Interactions option if required to control how slicers affect individual visuals.

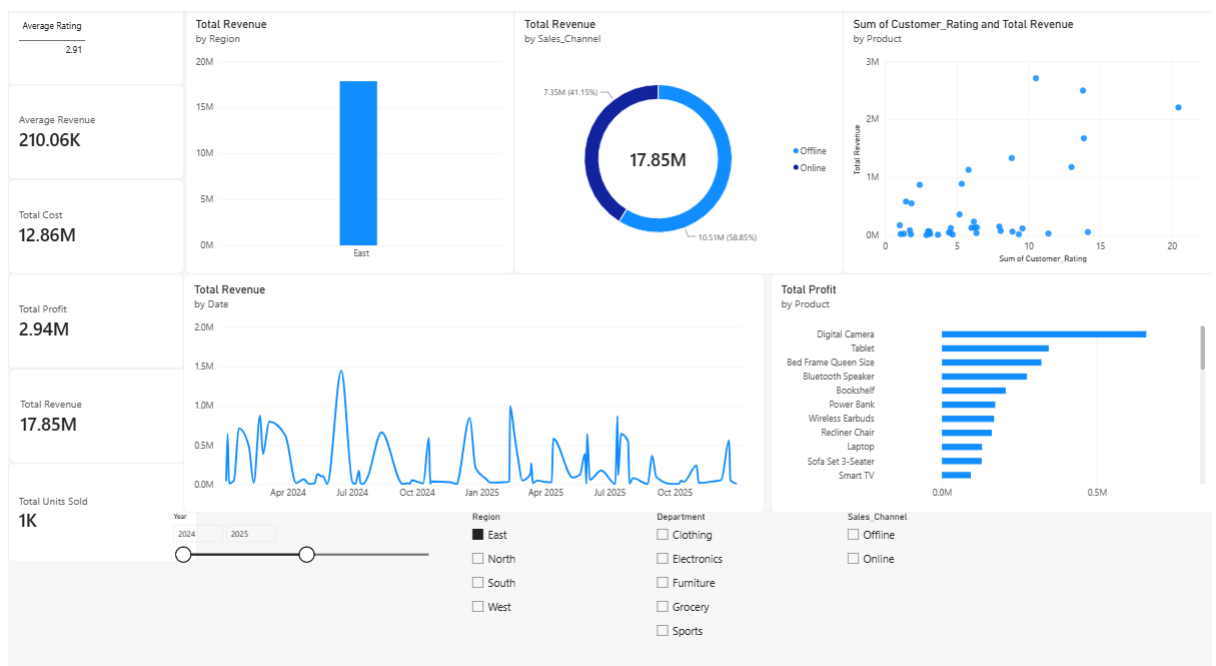
Output

1)Change in the charts according to the year slicer

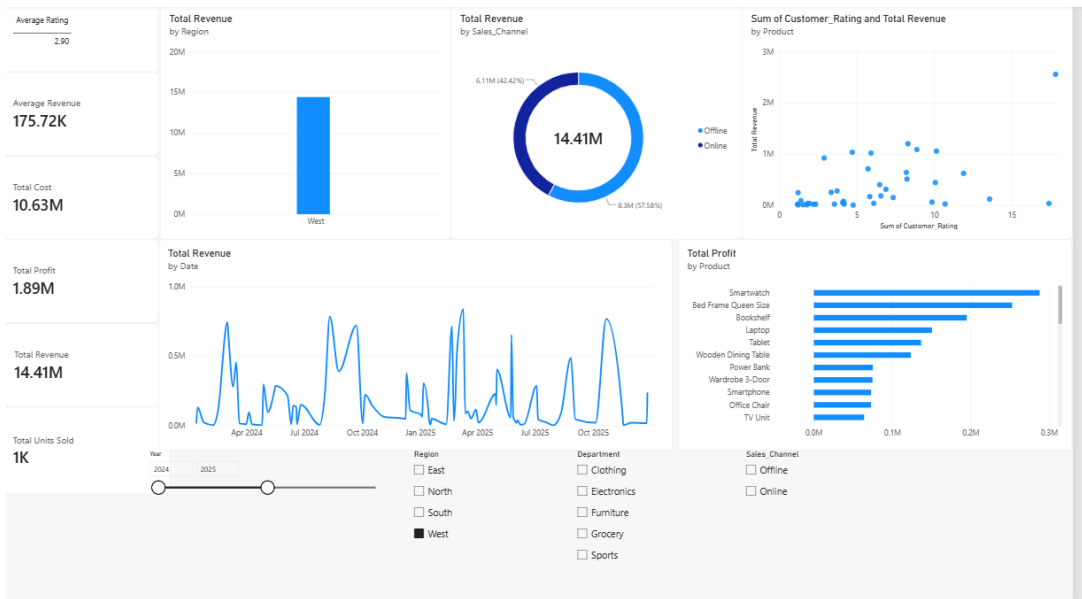


2) Change in charts according to the region slicer:

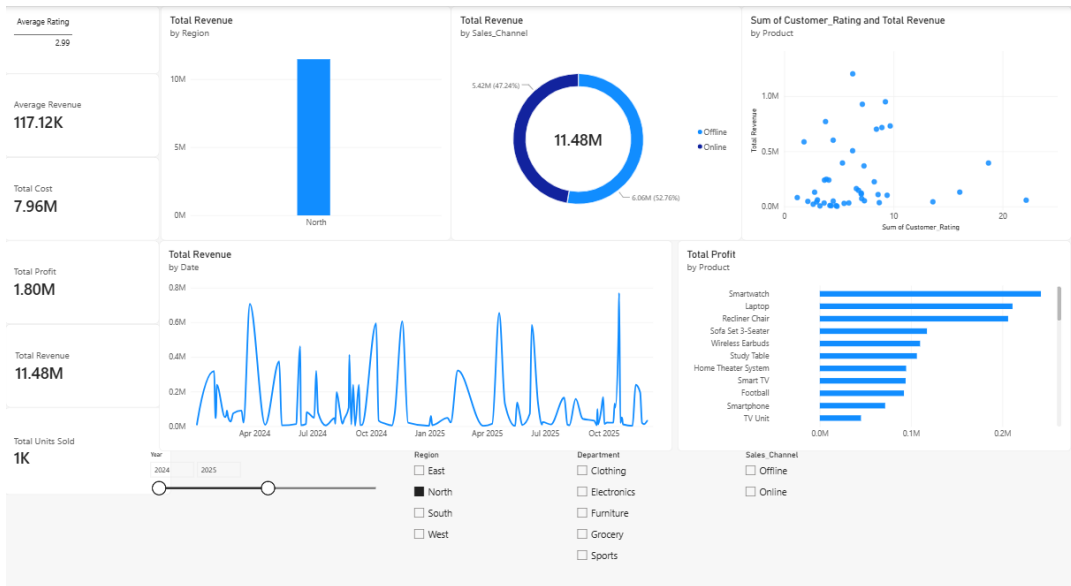
• East:



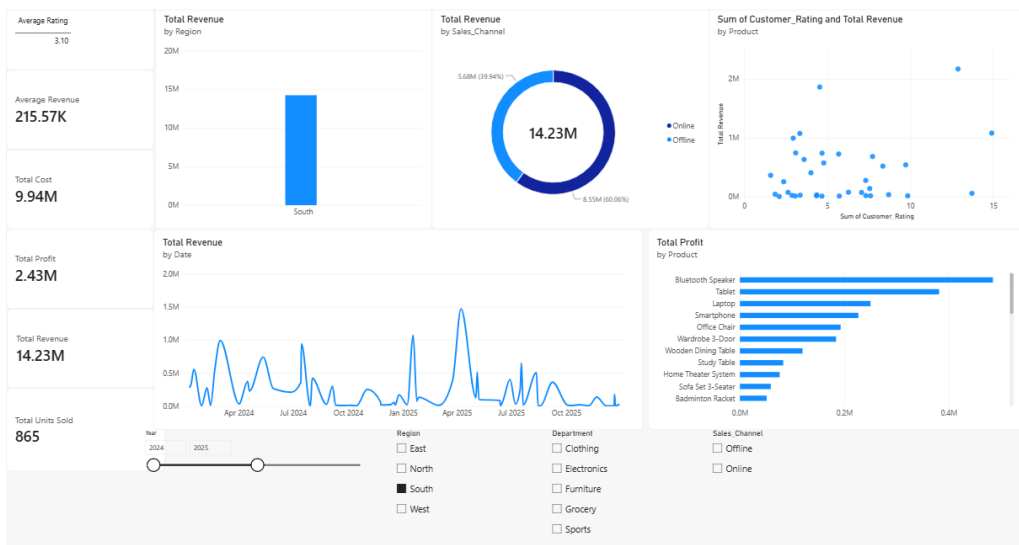
• West:



North:

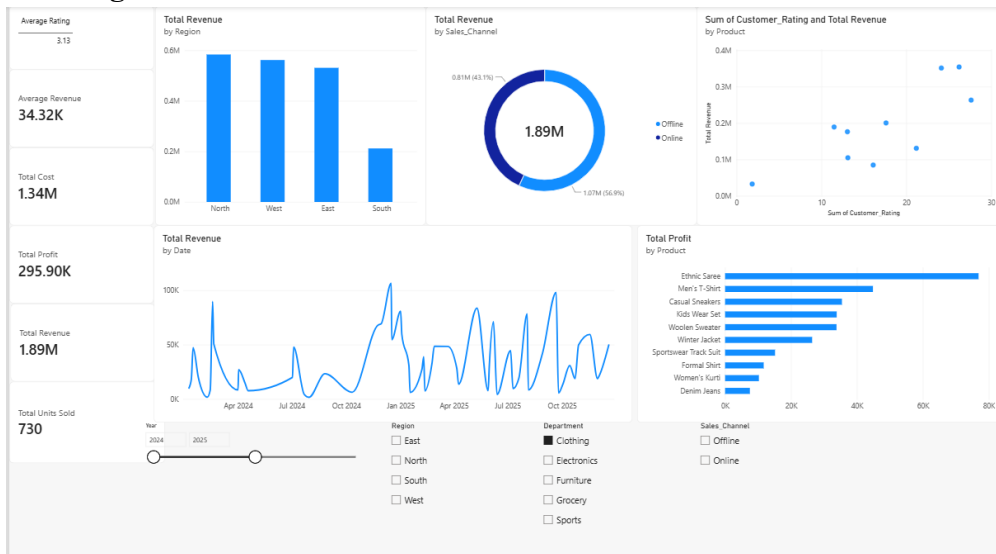


South:

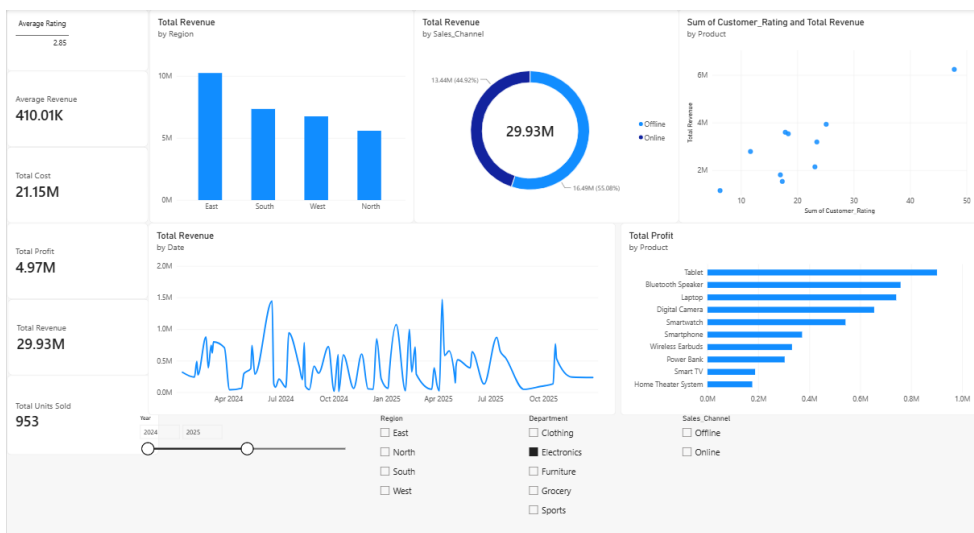


3)Change in charts according to the department slicer:

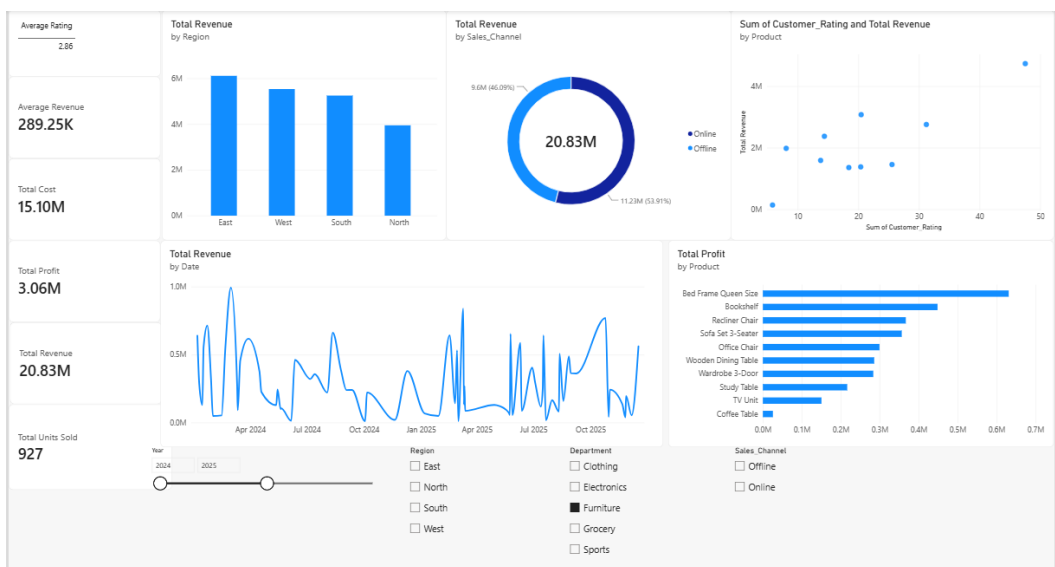
- **Clothing**



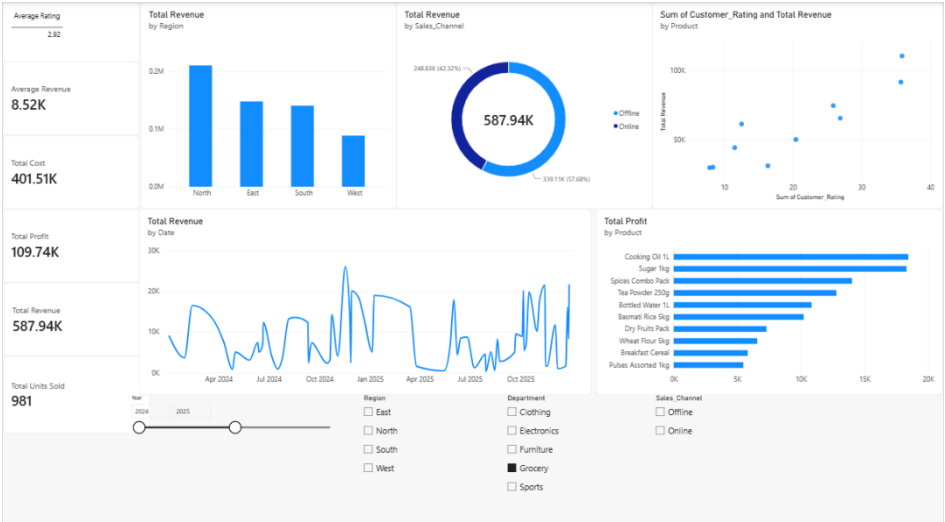
- **Electronics**



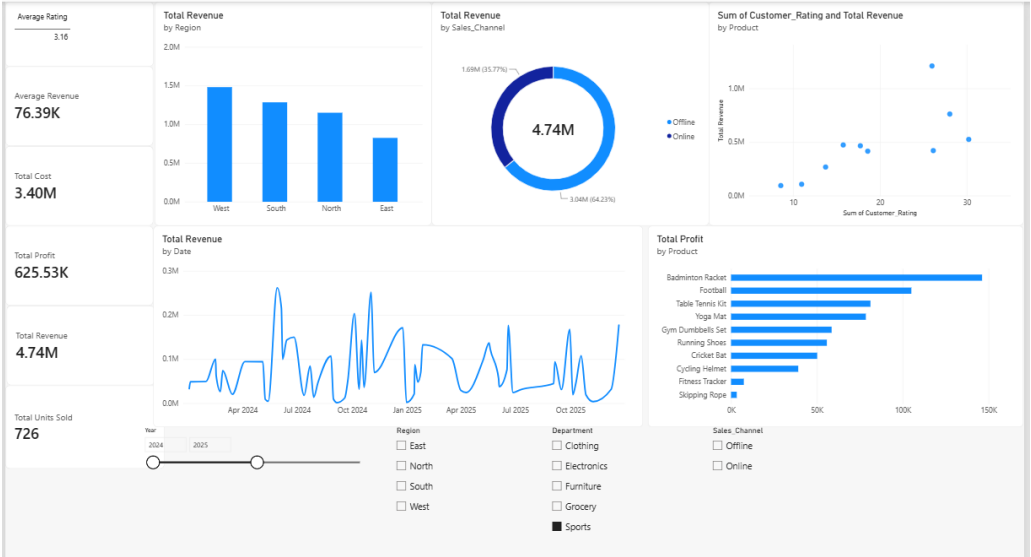
- **Furniture**



● Grocery

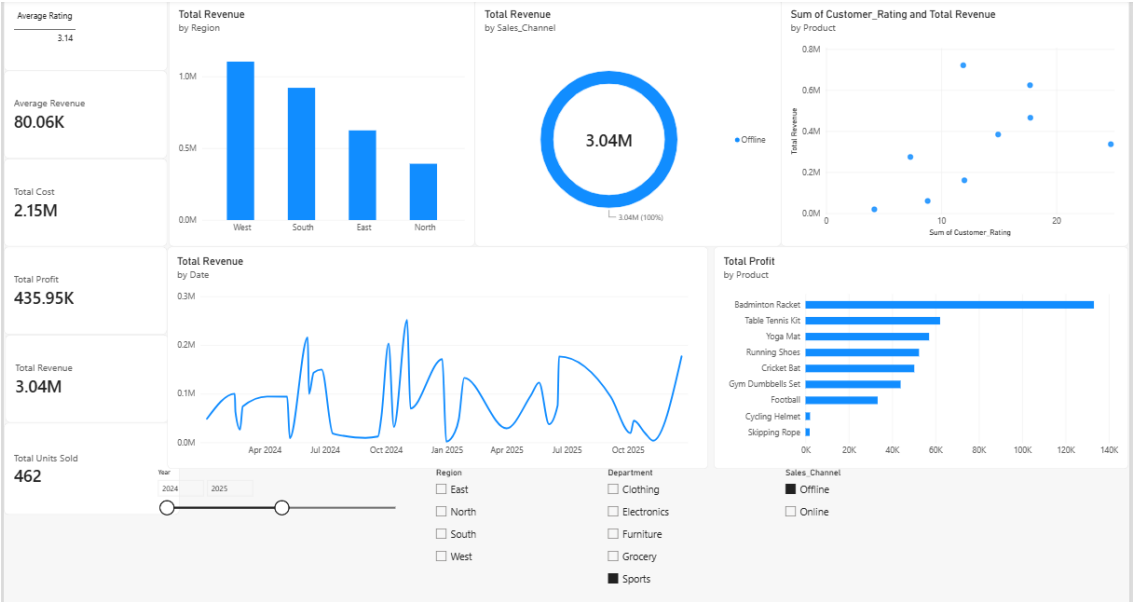


● Sports

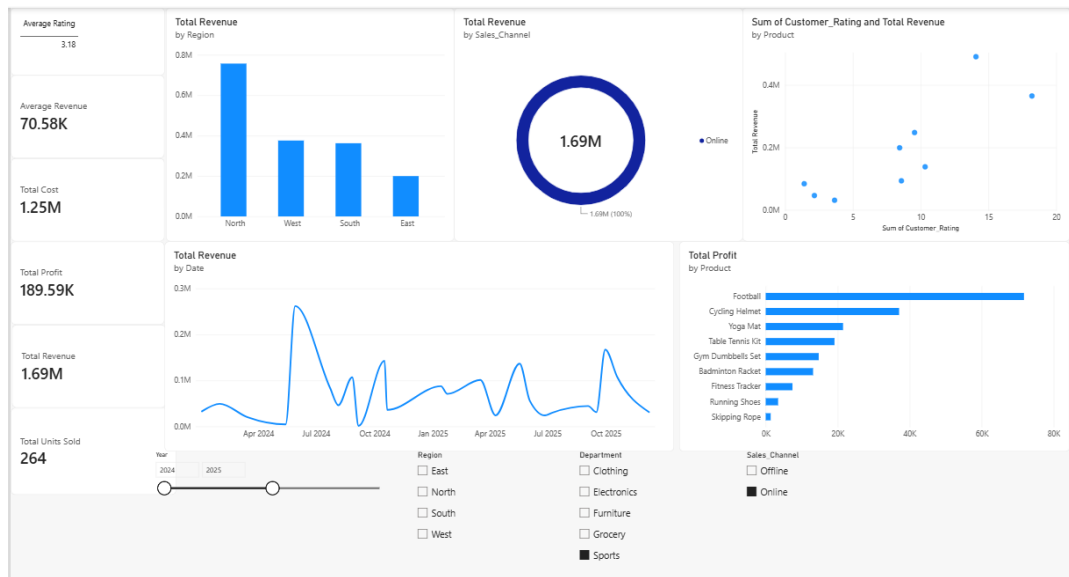


3)Changes according to the sales channel slicer:

● Offline:



- **Online**



9)FINAL DASHBOARD

Algorithm

1. Arrange all KPI cards, slicers, and visualizations in a structured layout.
2. Add a suitable dashboard title.
3. Apply consistent formatting to the visuals.
4. Enable data labels and meaningful chart titles.
5. Configure drill-down options for hierarchical analysis.
6. Create a drill-through page for detailed product-level analysis.
7. Enable visual interactions and cross-filtering.
8. Review the complete dashboard and verify its functionality.

Output

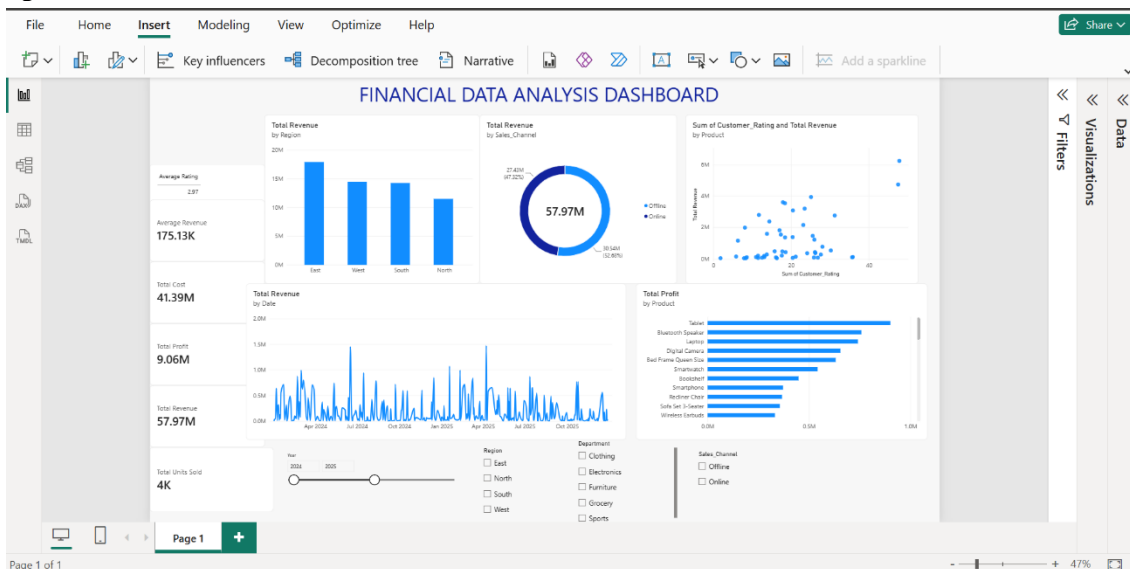


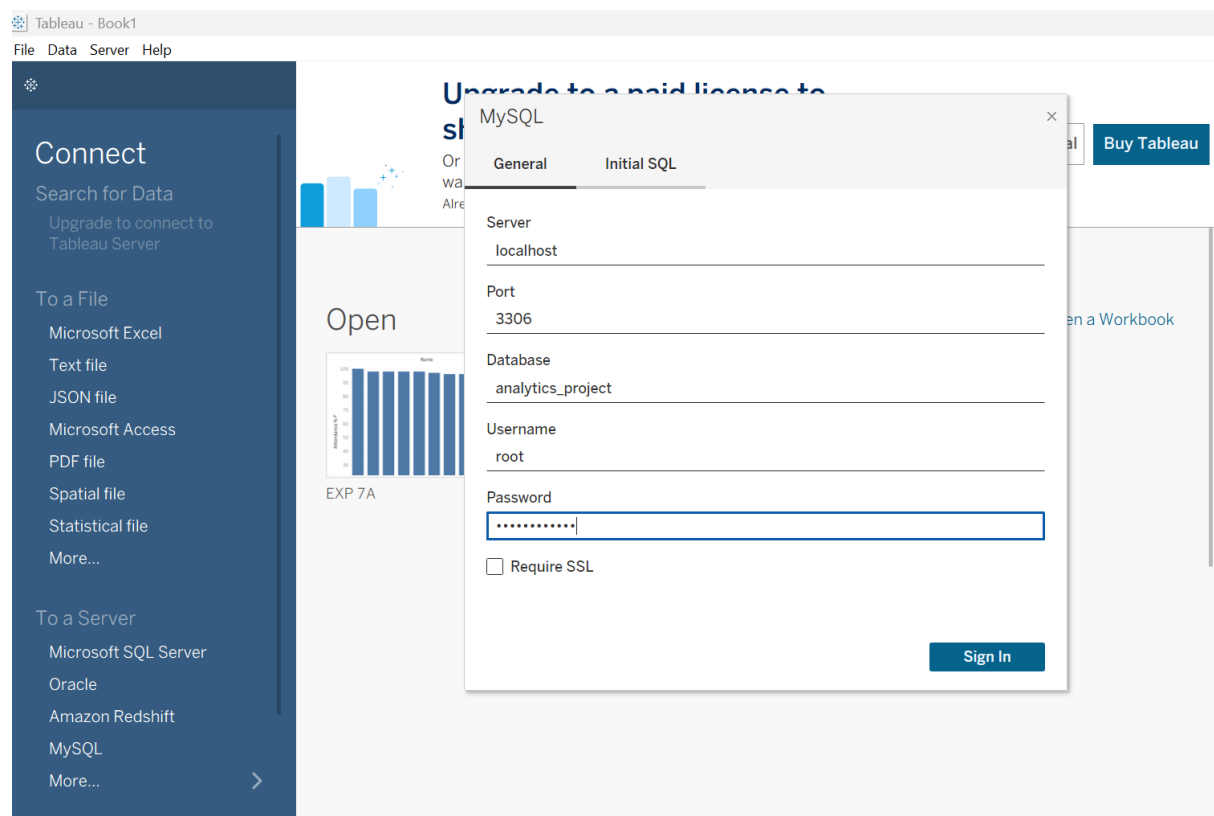
TABLEAU DASHBOARD:

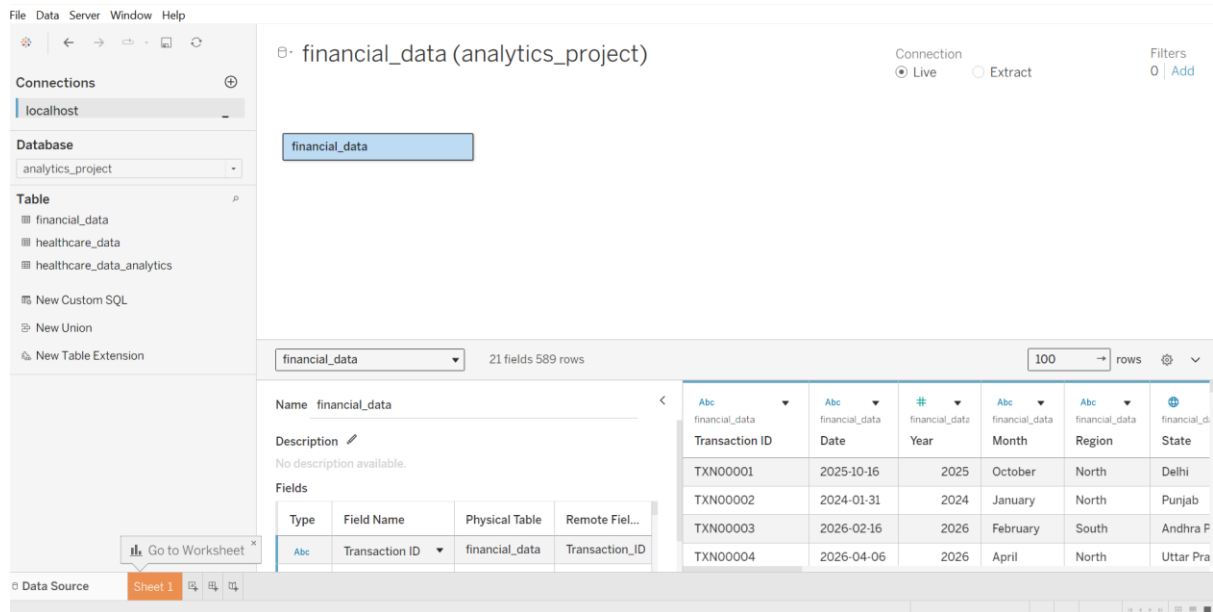
1)CONNECTION TO MYSQL SERVER:

Algorithm

1. Open Tableau Desktop and select the MySQL connector.
2. Enter the MySQL server and authentication details.
3. Select the database containing the required dataset.
4. Connect Tableau to the selected database.
5. Select the appropriate financial table.
6. Verify the data preview and field names.
7. Open the worksheet to begin data preparation and analysis.

Output





2)DATA PREPARATION AND INTEGRATION

Algorithm

1. Connect the financial_data_analytics table from MySQL to Tableau.
2. Verify the data types of date, numerical, and categorical fields.
3. Check the data for missing or inconsistent values and make the required corrections.
4. Organize the financial fields such as Revenue, Cost, Profit, Region, Product, and Sales Channel for analysis.
5. Verify the prepared dataset and proceed to create worksheets.

Output

Datatype of Date column is changes to Date, nulls removed and duplicates removed

financial_data

21 fields 589 rows

100 rows

Name financial_data

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Fiel...
Abc	Transaction ID	financial_data	Transaction_ID
📅	Date	financial_data	Date
#	Year	financial_data	Year
Abc	Month	financial_data	Month
Abc	Region	financial_data	Region
🌐	State	financial_data	State
🌐	City	financial_data	City
Abc	Department	financial_data	Department
Abc	Product	financial_data	Product

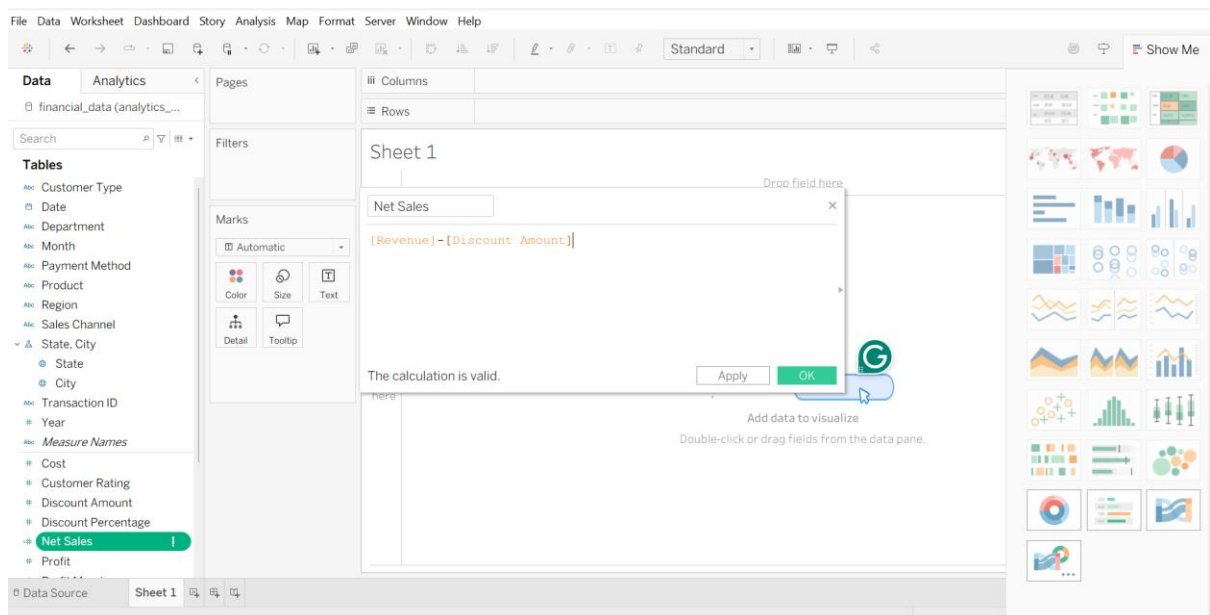
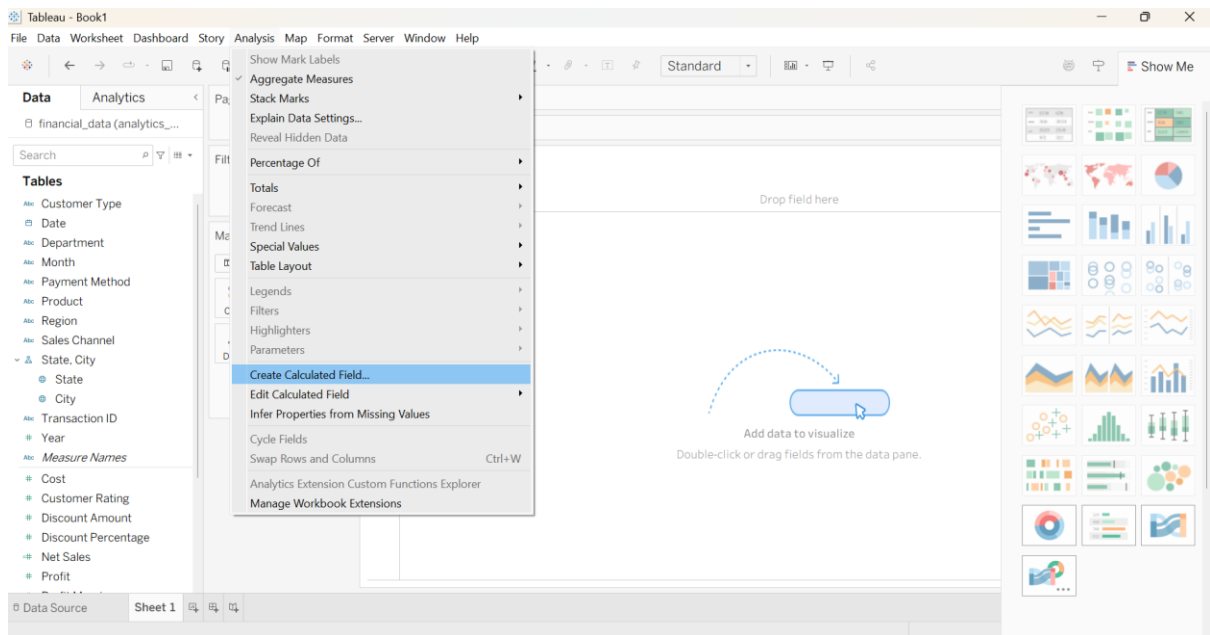
Abc	📅	#	Abc	Abc	🌐
financial_data	financial_data	financial_data	financial_data	financial_data	financial_data
Transaction ID	Date	Year	Month	Region	State
TXN00001	16-10-2025	2025	October	North	Delhi
TXN00002	31-01-2024	2024	January	North	Punjab
TXN00003	16-02-2026	2026	February	South	Andhra P
TXN00004	06-04-2026	2026	April	North	Uttar Pra
TXN00005	02-03-2026	2026	March	East	West Ber
TXN00006	30-11-2025	2025	November	North	Rajastha
TXN00007	28-11-2024	2024	November	North	Punjab
TXN00008	24-06-2026	2026	June	West	Gujarat
TXN00009	18-02-2024	2024	February	North	Punjab
TXN00010	23-08-2026	2026	August	North	Delhi
TXN00011	26-09-2024	2024	September	South	Telangana
TXN00012	15-05-2025	2025	May	North	Delhi

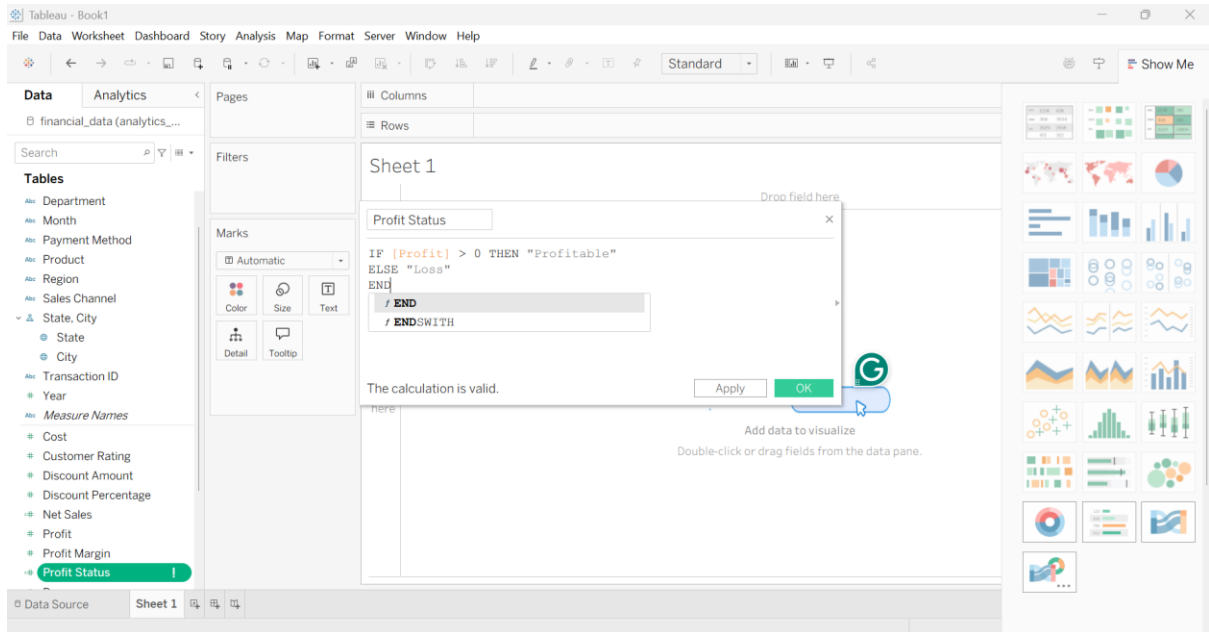
3)CALCULATED FIELDS

Algorithm

1. Create a calculated field named Net Sales.
2. Calculate net sales by subtracting the discount amount from revenue.
3. Create another calculated field named Profit Status.
4. Classify transactions as Profitable or Loss based on their profit value.
5. Verify the calculated fields and use them in subsequent visualizations.

Output



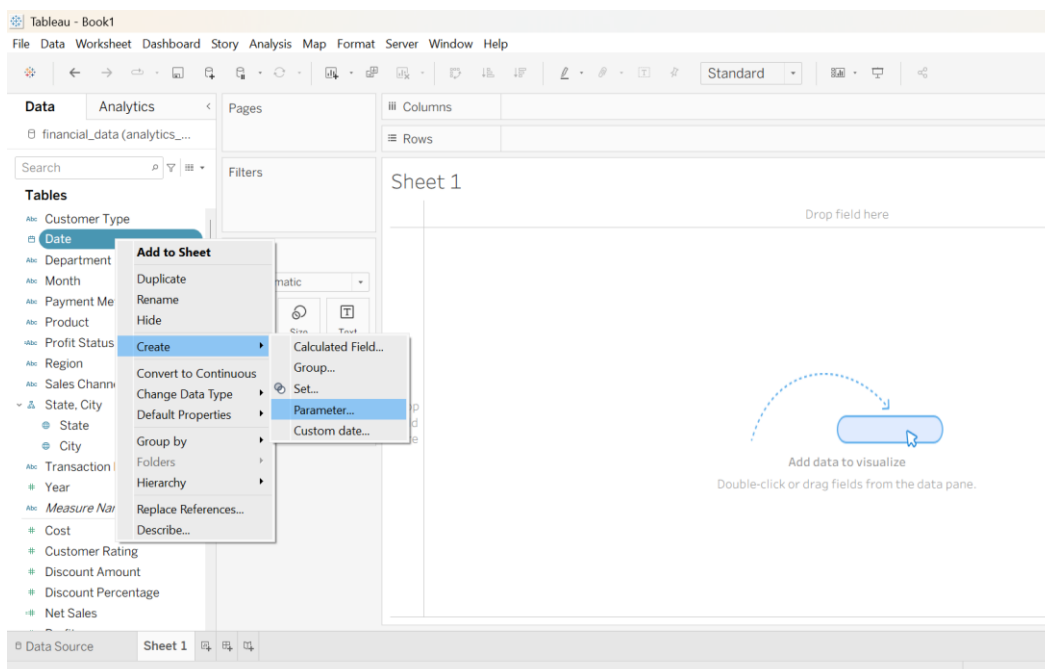


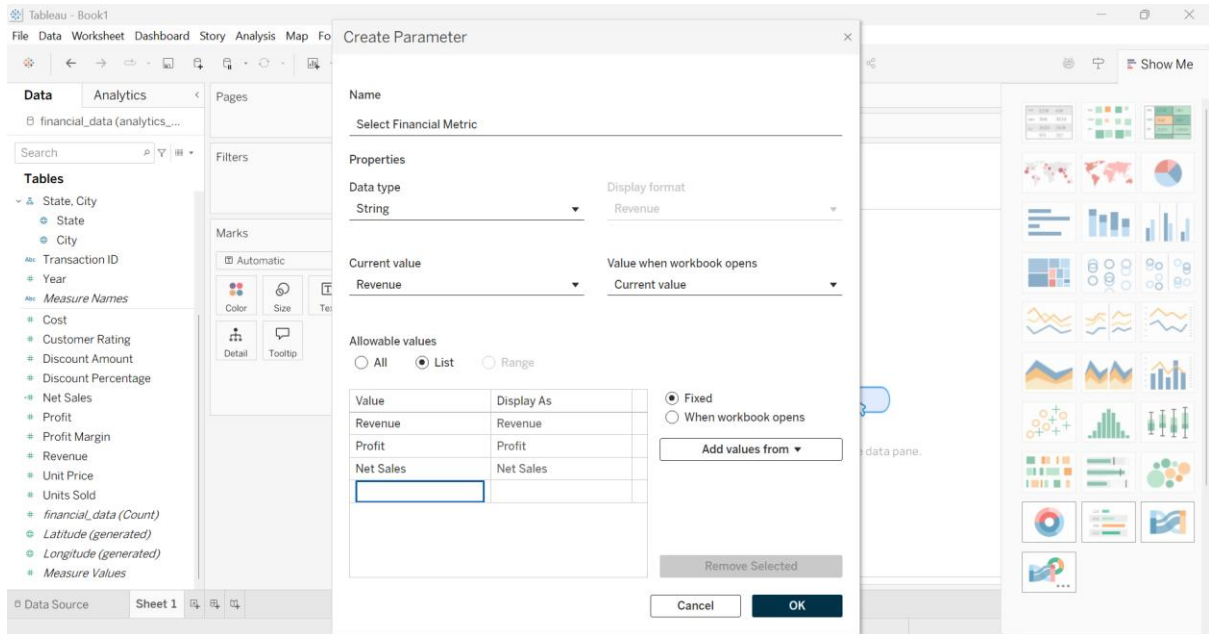
4)PARAMETERS

Algorithm

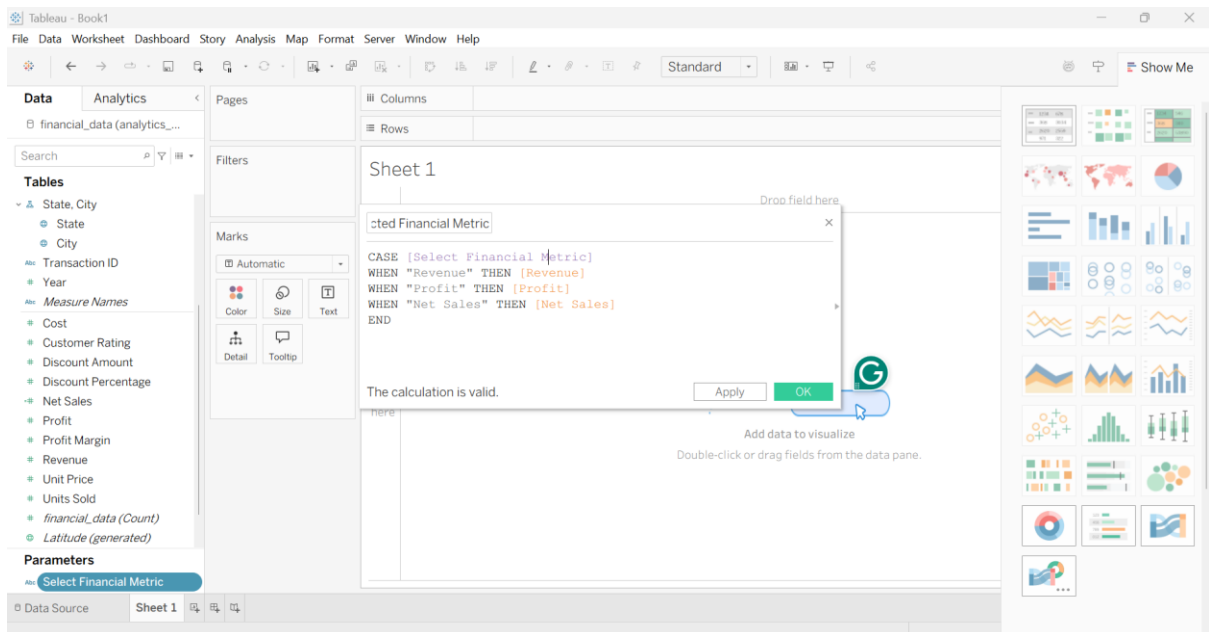
1. Create a parameter named Select Financial Metric.
2. Set its data type to String.
3. Add the values Revenue, Profit, and Net Sales.
4. Create a calculated field that displays the metric selected by the parameter.
5. Use the calculated field in a visualization to make it interactive.

Output





Creating the calculated field for this parameter



5) CREATING VISUALIZATIONS

Algorithm

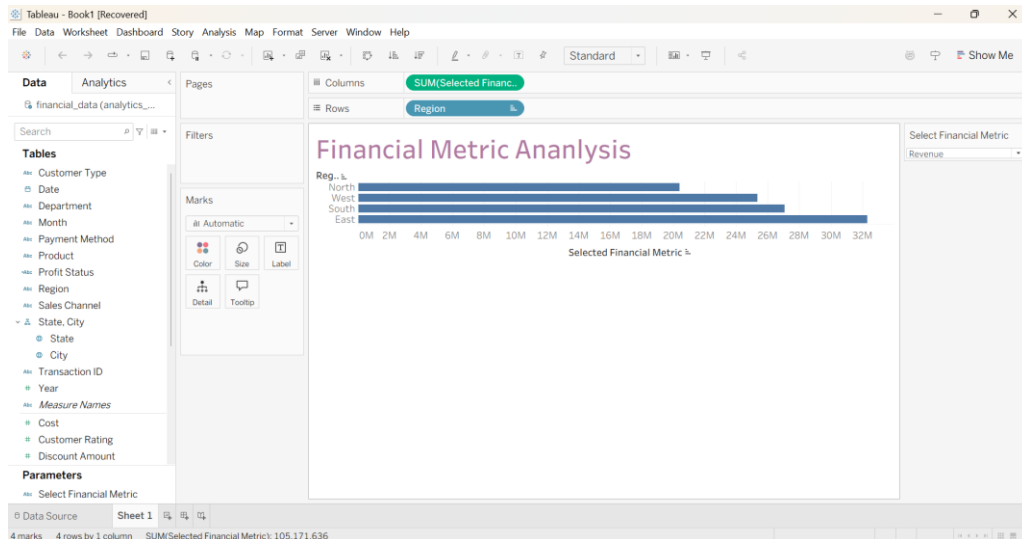
1. Create worksheets using appropriate dimensions and measures from the financial dataset.
2. Use the Select Financial Metric parameter to dynamically compare Revenue, Profit, and Net Sales.
3. Create a treemap to analyze departmental profitability and a highlight table to compare regional and product performance.

4. Create a geographical map to visualize profit across states.
5. Create an area chart to identify monthly changes in net sales.

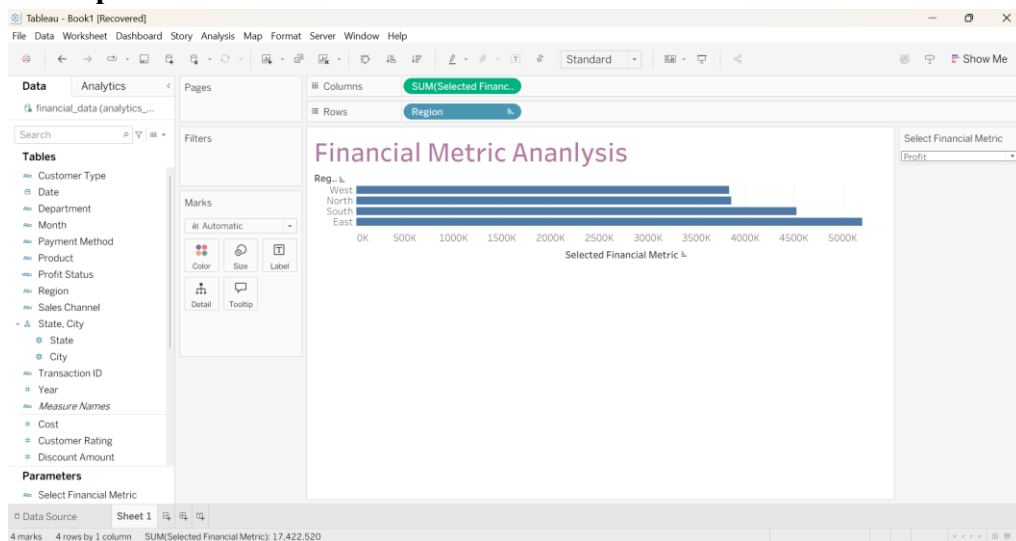
Output

1)Sheet 1 – Parameter-driven bar chart:

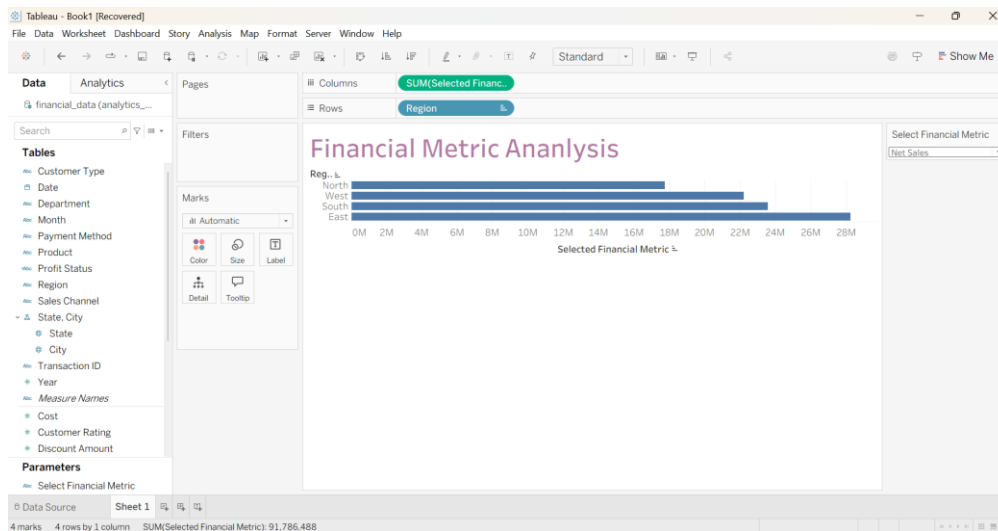
- **Revenue parameter:**



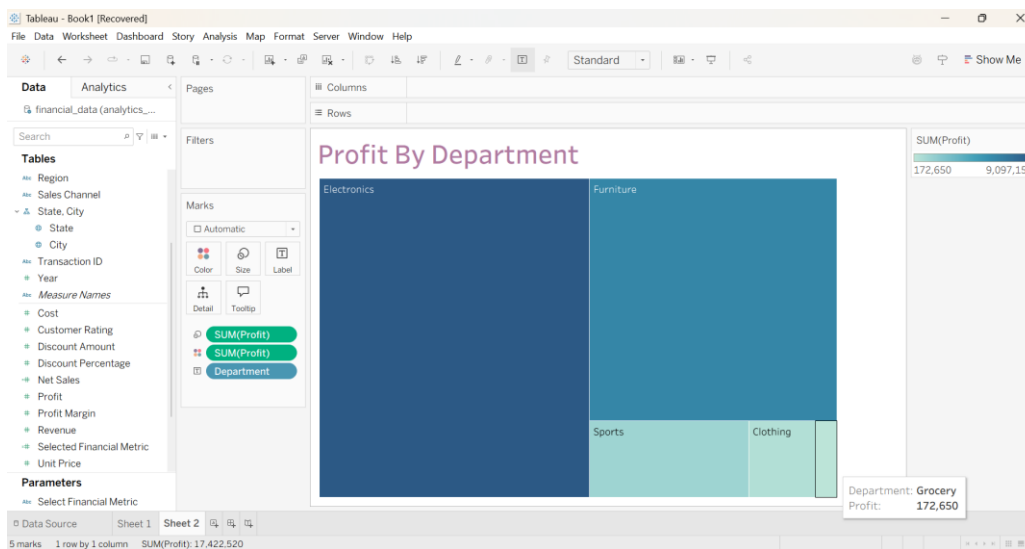
- **Profit parameter:**



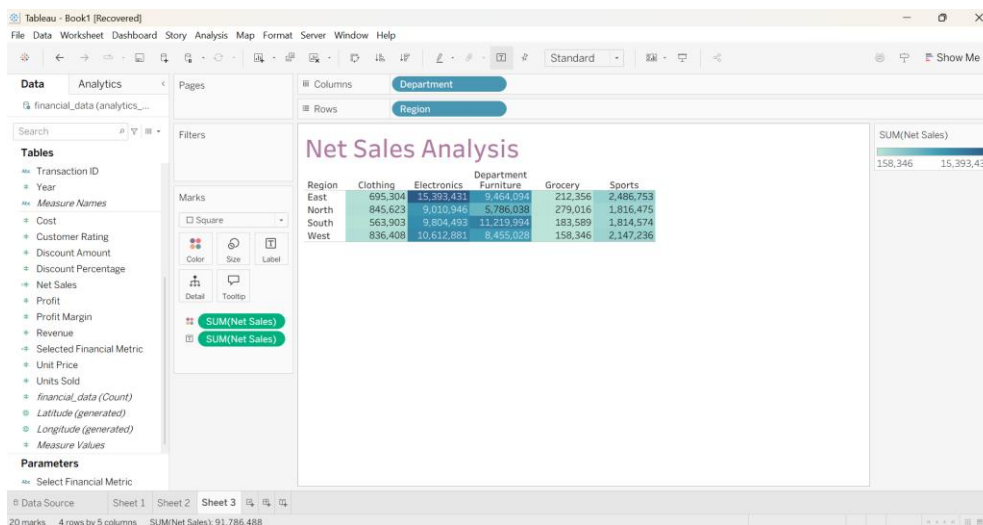
- **Net Sales parameter:**



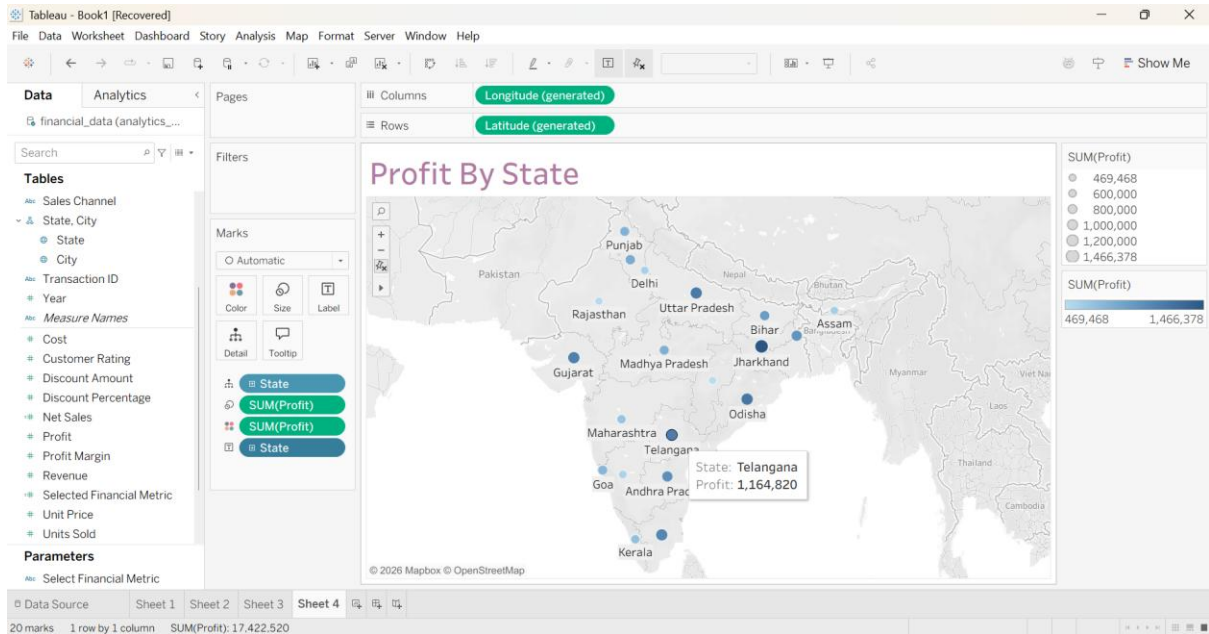
2)Sheet 2 – Profit by department:



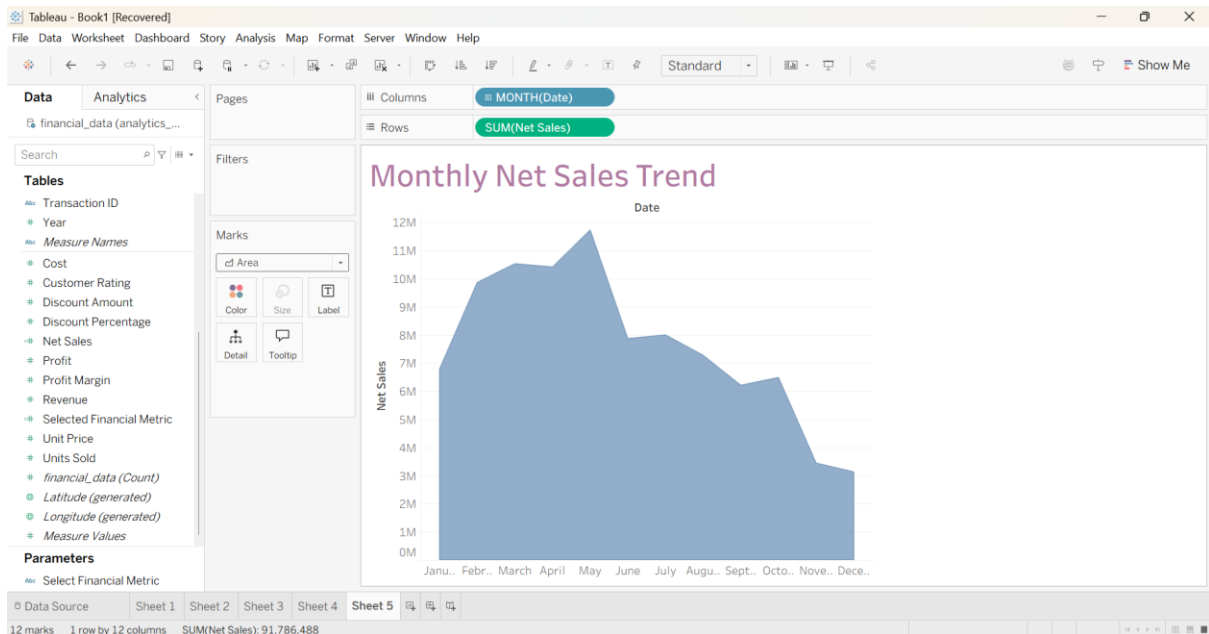
3)Sheet 3 – Highlight table:



4)Sheet 4 – Profit by state:



5)Sheet 5 – Monthly net sales trend:

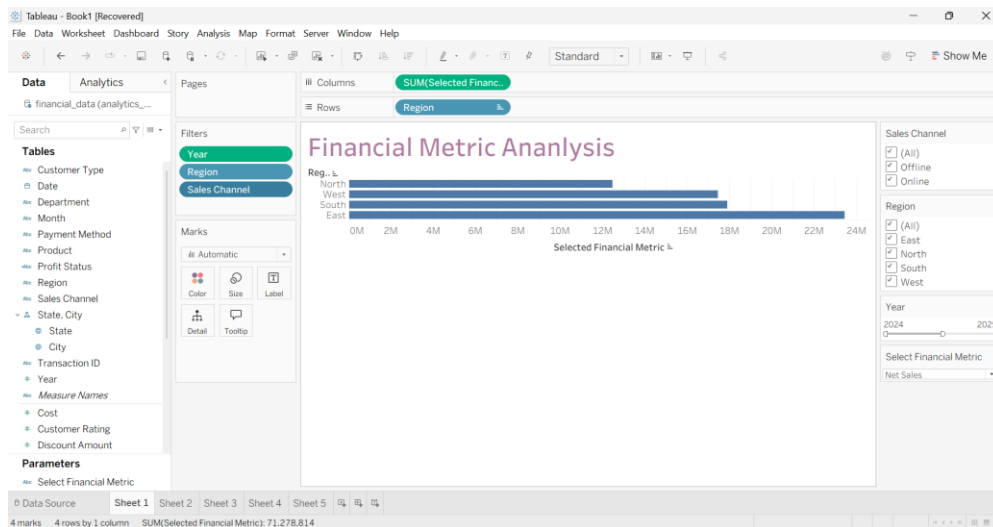


6)FILTERS

Algorithm

1. Add Year, Region, and Sales Channel as filters.
2. Select the required values from the filter controls.
3. Apply the filters to analyze financial data for specific periods, regions, or sales channels.

Output

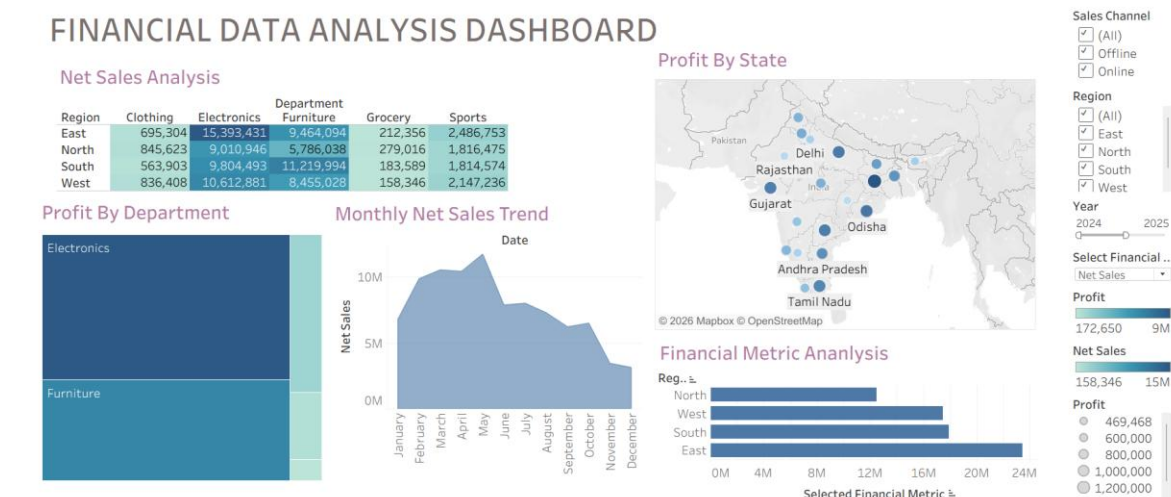


7) CREATING DASHBOARD

Algorithm

- 1) Create a new dashboard and add the five financial worksheets.
- 2) Arrange the visualizations in a structured layout.
- 3) Add the required filters and parameter control.
- 4) Configure the dashboard for interactive financial analysis.

Output



RESULT:

The financial dataset was successfully connected to MySQL and analyzed using Tableau. Created calculated fields, parameters, filters, and interactive visualizations to analyze revenue, profit, net sales, departmental performance, regional performance, and monthly trends. The resulting dashboard clearly visualizes key financial patterns and performance.